

Problem Descriptions

- 1. Randomized drone inspection angle.** A university research drone is used to inspect a circular construction site. During each inspection, its camera points in a direction D measured in radians clockwise from a fixed reference marker, with $D \in [0, 2\pi)$. Because the drone controller randomly selects the initial orientation with no preferred direction, a circular-uniform model is appropriate. The direction 2π is equivalent to 0 because they represent the same physical orientation.

The construction team is particularly interested in directions close to the fixed reference marker. Determine the complete periodic probability density rule for the camera direction, including what happens outside the stated cycle. Then find the probability that the camera direction lies within $\frac{\pi}{4}$ radians of the reference marker, interpreting the interval around 0 periodically.
- 2. Periodic triangular energy-demand time.** A university campus monitors the time T at which its electricity demand reaches its daily peak. The time is measured in hours on $[0, 24)$, with midnight represented by both 0 h and 24 h. Therefore, time of day is periodic with period 24 h.

For this planning model, the electricity demand peak is least likely at midnight. The density increases linearly from zero at midnight to its maximum at 10:00, then decreases linearly back to zero at the next midnight. A circular-triangular pdf represents this daily pattern.

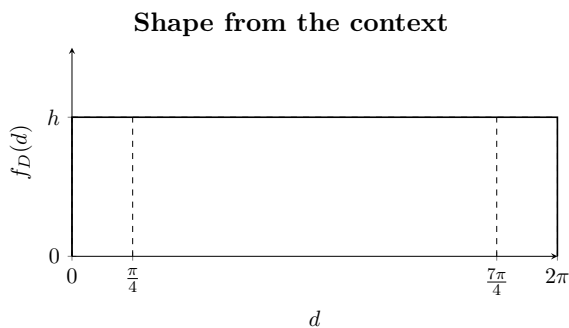
The campus energy team is interested in peak times close to midnight. Determine the complete periodic probability density rule for the peak time, including what happens outside the displayed day. Then find the probability that the daily peak occurs within 5 hours of midnight, interpreting the interval around midnight periodically.

Solutions

1 Periodic uniform distribution

1.1 Support and Shape

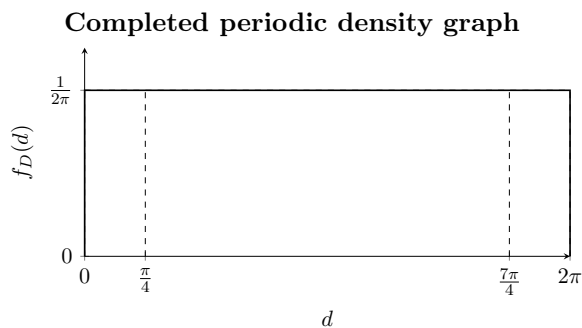
The random variable D represents the drone camera direction in radians. One complete cycle is represented by $[0, 2\pi)$, and the direction is periodic because a full rotation returns the camera to the same physical orientation. The controller selects all directions with equal likelihood, so the density is a rectangle of unknown height h over the complete cycle and is zero outside the displayed support.



1.2 Total Area = 1

The rectangular pdf must have total area 1 over one complete cycle:

$$1 = (2\pi)h, \quad h = \frac{1}{2\pi}.$$



1.3 Straight Line Rule

The density is constant throughout the support, so the graph is horizontal. Therefore its slope is $m = 0$, and the constant rule is $f_D(d) = \frac{1}{2\pi}$ on the support. The periodic nature of the direction is expressed by $f_D(d + 2\pi) = f_D(d)$.

1.4 Complete pdf

The complete pdf can be written on one displayed cycle as

$$f_D(d) = \begin{cases} \frac{1}{2\pi}, & 0 \leq d < 2\pi, \\ 0, & \text{otherwise.} \end{cases}$$

The displayed endpoints 0 and 2π represent the same physical direction, and the periodic extension satisfies

$$f_D(d + 2\pi) = f_D(d).$$

1.5 Interval Around the Reference Marker

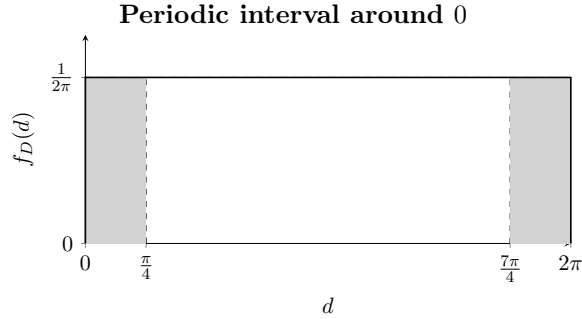
The interval is

$$-\frac{\pi}{4} \leq D \leq \frac{\pi}{4} \quad \longrightarrow \quad \left[0, \frac{\pi}{4}\right] \cup \left[\frac{7\pi}{4}, 2\pi\right).$$

The two pieces have total width

$$\frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \quad \longrightarrow \quad P\left(-\frac{\pi}{4} \leq D \leq \frac{\pi}{4}\right) = \frac{\pi}{2} \frac{1}{2\pi} = \frac{1}{4} = 25\%.$$

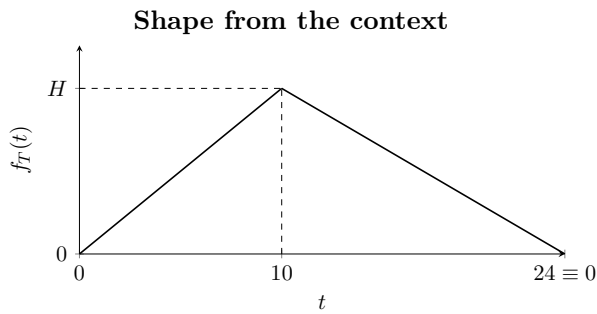
Thus, 25% of randomized camera directions lie within $\frac{\pi}{4}$ radians of the reference marker.



2 Periodic triangular distribution

2.1 Support and Shape

The random variable T represents the time of the daily electricity-demand peak in hours. One complete cycle is represented by $[0, 24)$, and the period is 24 h because the clock repeats every day. The density is zero at midnight, rises steadily to its maximum at 10 h, and then falls steadily to zero at the next midnight. Therefore, the graph is a triangle with unknown maximum height H and mode at $t = 10$.

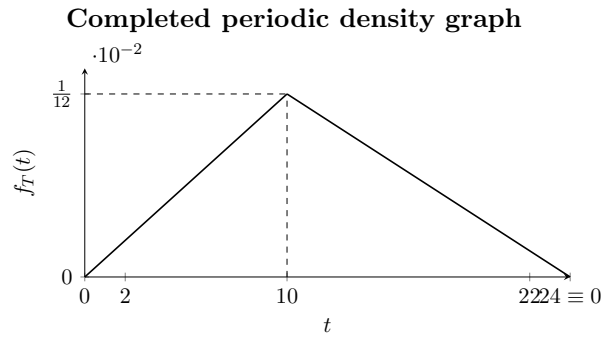


2.2 Total Area = 1

The triangular pdf must have total area 1:

$$1 = \frac{1}{2}(24)H, \quad H = \frac{1}{12}.$$

The completed triangle therefore has maximum height $1/12$ at $t = 10$.



2.3 Straight Line Rules

The rising section connects $(0, 0)$ and $(10, \frac{1}{12})$:

$$(0, 0), \left(10, \frac{1}{12}\right) \quad \longrightarrow \quad m = \frac{\frac{1}{12} - 0}{10 - 0} = \frac{1}{120}, \quad b = 0 \quad \longrightarrow \quad f_T(t) = \frac{t}{120}.$$

The falling section connects $(10, \frac{1}{12})$ and $(24, 0)$:

$$\left(10, \frac{1}{12}\right), (24, 0) \quad \longrightarrow \quad m = \frac{0 - \frac{1}{12}}{24 - 10} = -\frac{1}{168} \quad \longrightarrow \quad f_T(t) = \frac{24 - t}{168}.$$

2.4 Complete pdf

The complete pdf on one displayed day is

$$f_T(t) = \begin{cases} \frac{t}{120}, & 0 \leq t \leq 10, \\ \frac{24 - t}{168}, & 10 < t < 24, \\ 0, & \text{otherwise.} \end{cases}$$

The periodic extension is

$$f_T(t + 24) = f_T(t).$$

The two triangular sections have areas

$$\frac{1}{2}(10)\frac{1}{12} + \frac{1}{2}(14)\frac{1}{12} = \frac{5}{12} + \frac{7}{12} = 1,$$

confirming that the pdf is correctly normalized.

2.5 Interval Around Midnight

The interval is

$$-5 \leq T \leq 5 \quad \longrightarrow \quad [0, 5] \cup [19, 24).$$

For $[0, 5]$:

$$f_T(5) = \frac{5}{120} = \frac{1}{24} \quad \longrightarrow \quad A_1 = \frac{1}{2}(5)\frac{1}{24} = \frac{5}{48}.$$

For $[19, 24)$:

$$f_T(19) = \frac{24 - 19}{168} = \frac{5}{168} \quad \longrightarrow \quad A_2 = \frac{1}{2}(5)\frac{5}{168} = \frac{25}{336}.$$

Therefore,

$$P(-5 \leq T \leq 5) = \frac{5}{48} + \frac{25}{336} = \frac{35}{336} + \frac{25}{336} = \frac{60}{336} = \frac{5}{28} \approx 17.86\%.$$

Thus, approximately 17.9% of days have their modeled electricity-demand peak within 5 hours of midnight.

